Multivariate Machine Learning for Cyber-Hygiene Anomaly Detection Across Active Directory, Endpoint, and Patch Telemetry

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***Abstract*: Security operations teams watch endpoint fleets through three telemetry channels: Active Directory, endpoint posture, and patch state. A compromised or misconfigured host appears as an anomaly in this telemetry, and the operational question is whether multivariate machine-learning anomaly detection earns its complexity over the per-signal threshold rules teams already run. We answer with a pre-registered, fully reproducible benchmark of 1,500 hosts with a 12-dimensional hygiene telemetry vector, injecting two anomaly types: a single-channel anomaly that crosses a per-feature threshold, and a cross-channel anomaly that keeps every feature individually normal but breaks the correlation structure normal hosts exhibit. The answer is conditional. On cross-channel anomalies a covariance-aware detector reaches average precision of 0.710 against 0.103 for the best per-feature rule (difference 0.607; 95% BCa interval $[0.598, 0.617]$). On single-channel anomalies the rule wins, 0.836 against 0.740 for the best joint detector. The cross-channel advantage requires joint modeling, not aggregation: a structure-blind exceedance count reaches 0.084. The choice of method is decisive: covariance-aware (Mahalanobis, 0.710) and density-aware (Local Outlier Factor, 0.638) detectors see the violation, while the axis-aligned Isolation Forest does not (0.105). On a fifty-fifty mixture the joint detector still leads (0.702 against 0.473) but by a smaller margin (0.2285) than on cross-channel anomalies alone. The deployable conclusion is a two-tier monitor: cheap rules for the obvious single-channel spikes and a covariance- or density-aware detector for the subtle cross-channel violations. The popular Isolation Forest is the wrong default for this anomaly class. We supply a closed-form account: because the cross-channel anomaly preserves every marginal, per-feature and axis-aligned detectors are provably at the prevalence floor while the expected squared Mahalanobis distance separates the two covariance structures through the nonzero off-diagonal terms of the inverse covariance, and this same structure is what a correlation-only adversarial evasion exploits.**

***Index Terms*: anomaly detection, cyber hygiene, Active Directory, endpoint telemetry, Isolation Forest, Local Outlier Factor, Mahalanobis distance, average precision, pre-registration**

I. Introduction

A SECURITY operations team monitoring a large endpoint fleet sees three telemetry channels. The first is Active Directory: privileged logon counts, security-group churn, the age of stale administrative accounts, and the rate of failed authentications. The second is endpoint posture: process-spawn anomalies, the fraction of unsigned binaries executed, configuration drift from a hardened baseline, and host-firewall state. The third is patch state: aggregate patch debt, days since the last successful patch, the presence of known-exploited vulnerabilities [1], and the staleness of the host's own telemetry. Each channel is a small bundle of numbers reported per host, and a fleet of a few thousand hosts produces a continuous stream of these vectors. A host that has been compromised, misconfigured, or left to rot shows up somewhere in this stream as an anomaly.

Teams already run per-signal threshold rules over these channels, because rules are cheap, auditable, and easy to explain to an authorizing official. A rule fires when patch debt crosses a line, when an account has been dormant past a deadline, or when too many unsigned binaries run in a day. Rules of this kind catch the blatant cases and form the backbone of most operational hygiene dashboards. The question this paper asks is whether multivariate machine-learning anomaly detection [2], [3] earns its additional complexity over those rules, and if so, for which anomalies. The complexity is real: a joint detector must be fit, maintained, explained, and trusted, and the anomaly-detection literature offers a confusing menu of methods with no consensus default. A team that adopts the wrong method, or adopts any method where a rule would have done, pays for the machinery without the benefit.

Our hypothesis, fixed before any evaluation, is that the answer depends on the geometric structure of the anomaly rather than on a blanket judgment about machine learning. We distinguish two anomaly types. A *single-dimension anomaly* (SDA) is a threshold-crossing spike: one channel shifts hard, patch debt jumps, an account count explodes, and a per-feature rule catches it by construction. A *cross-dimension anomaly* (CDA) is subtler. Every feature stays individually within its normal range, so no per-feature rule fires, but the correlation structure that normal hosts exhibit is broken. Normal hosts have channels that move together in characteristic ways; a host whose channels have become mutually inconsistent, each plausible alone but jointly impossible, is exactly the compromise-like case that per-feature inspection cannot see and a joint detector should. If this distinction governs the outcome, the practical guidance is specific rather than ideological: deploy multivariate detection for the subtle cross-channel cases and keep cheap rules for the obvious single-channel ones.

We answer with a pre-registered, reproducible benchmark. A controlled fleet of 1,500 hosts carries a 12-dimensional hygiene vector with documented within-channel correlation. Two anomaly types are injected at fixed prevalence, and three evaluation conditions, single-channel only, cross-channel only, and a fifty-fifty mixture, run on the same seeds. Six detectors compete: two per-feature rule baselines and four joint detectors drawn from the standard anomaly-detection toolbox, all held at fixed, untuned settings so that no detector gains from threshold or hyperparameter search. Evaluation is threshold-free, by average precision [4], the appropriate metric for ranking rare anomalies, with verdicts decided by the non-overlap of bias-corrected and accelerated (BCa) bootstrap intervals [5], [6] rather than by significance tests. The methodology follows the pre-registration discipline now standard in empirical software engineering [7]: hypotheses, thresholds, and the evaluation seed range were locked before any evaluation result was inspected.

The contribution is fourfold.

- We show that the value of multivariate detection is *conditional on anomaly structure*, not absolute. On cross-channel anomalies the best joint detector reaches average precision 0.710 against 0.103 for the best rule, a difference of 0.607 ($[0.598, 0.617]$); on single-channel anomalies the rule wins, 0.836 against 0.740. Treating the two cases the same is the central error.
- We show the cross-channel gain *requires joint modeling, not aggregation*. A structure-blind detector that merely counts per-feature exceedances reaches 0.084 on cross-channel anomalies against 0.710 for the covariance-aware detector (difference 0.626). Summing marginal alarms does not recover a correlation violation.
- We show the *choice of joint method is decisive*. Covariance-aware (Mahalanobis, 0.710) and density-aware (Local Outlier Factor, 0.638) detectors see the violation; the axis-aligned Isolation Forest (0.105) does not. A team that reaches for the most popular library detector would wrongly conclude that machine learning does not help here.
- We translate these into a deployable *two-tier monitor*: cheap per-feature rules for the obvious single-channel spikes and a covariance- or density-aware detector for the subtle cross-channel violations, with the explicit warning that the default Isolation Forest is the wrong tool for this anomaly class.

II. Background and Related Work

A. Anomaly detection: taxonomy and surveys

Anomaly detection is a mature field with two influential surveys that frame the present problem. Chandola, Banerjee, and Kumar [2] organize the field by the nature of the anomaly (point, contextual, and collective), by whether labels are available, and by output type (label versus score), and they emphasize that the right method depends on the structure of the data and of the anomaly rather than on a universal best algorithm. Hodge and Austin [3] survey outlier-detection methodologies across statistics, machine learning, and neural networks and make the same point from the statistical side: a method tuned to one kind of deviation can be blind to another. Our two anomaly types are both point anomalies in their taxonomy, but they differ in geometry: the single-channel anomaly is a large marginal deviation that any univariate method sees, while the cross-channel anomaly is a deviation only in the joint distribution, invisible to any method that inspects features one at a time. The surveys predict exactly the conditional outcome we test, that the appropriate detector tracks the geometry of the anomaly.

B. Density, boundary, and isolation methods

The joint detectors we compare span the three dominant algorithmic families, and their differences are the heart of the method-choice result. *Density-based* methods score a point by how isolated it is relative to the local density of its neighbors; the Local Outlier Factor (LOF) of Breunig et al. [8] compares a point's local reachability density to that of its neighbors, so a host that sits in a sparse region of the joint feature space, including a region made sparse because its channels are mutually inconsistent, receives a high score. *Boundary* methods estimate the support of the normal distribution and flag points outside it; the One-Class SVM of Schölkopf et al. [9] fits a kernel boundary around the bulk of the data, and its sensitivity to a correlation violation depends on whether the kernel and its bandwidth resolve the relevant geometry. *Isolation* methods isolate anomalies with random recursive partitioning; the Isolation Forest of Liu, Ting, and Zhou [10] builds trees of random *axis-aligned* splits and scores a point by how few splits isolate it. The axis-aligned construction is precisely why the Isolation Forest is expected to struggle on a correlation violation: a split on a single feature cannot, by itself, express that two features have become jointly inconsistent while each remains marginally normal. Outlier ensembles [11] combine such base detectors to hedge against any single method's blind spots; we deliberately evaluate the base detectors individually, because the purpose here is diagnostic, to expose which geometric assumption each family encodes, rather than to build the strongest possible combined detector.

C. Robust covariance and the Mahalanobis distance

The parametric counterpart to these methods is the Mahalanobis distance under a fitted covariance, the natural detector for a correlation-structure violation because it measures distance in the metric induced by the inverse covariance. A host whose channels violate the normal correlation lies far from the mean in this metric even when its per-feature values are unremarkable. The estimator matters: an ordinary sample covariance is itself sensitive to the anomalies it is meant to detect, so a robust covariance estimator is preferred. The Minimum Covariance Determinant estimator and its fast algorithm of Rousseeuw and Van Driessen [12] fit the covariance to the densest subset of the data, resisting contamination by outliers, and yield a Mahalanobis distance that is robust to the very anomalies present at evaluation time. This is the detector we expect to be best on cross-channel anomalies, because the cross-channel anomaly is, by construction, a covariance violation, and a covariance-aware metric is the one that represents it.

D. Why average precision for rare anomalies

Anomalies are rare, and rare-event ranking must be scored by a metric that is not dominated by the abundant normal class. Davis and Goadrich [4] show that the precision-recall curve, and its summary the average precision, is more informative than the ROC curve when the positive class is rare, because ROC area can look deceptively high when there are many easy true negatives. Average precision is also threshold-free: it summarizes the entire ranking a detector produces, which is what an operations team consumes when it triages hosts in score order, so no detector gains or loses from an arbitrary threshold choice. We therefore report average precision as the primary metric for every detector and condition, and we evaluate the same ranking with no per-detector threshold tuning.

E. Federal hygiene-monitoring context

Continuous fleet-wide hygiene monitoring is a federal expectation rather than an optional practice. FIPS 199 [13] sets the security categorization that determines the rigor a system owes, the NIST SP 800-53 control catalog [14] prescribes the configuration-management, audit, and identity controls whose telemetry our channels stand in for, and SP 800-137 [15] establishes information security continuous monitoring as an ongoing program. Sector policy such as the CJIS Security Policy [16] binds criminal-justice agencies to comparable hygiene and audit requirements. None of these documents prescribes a detection method; they require that hosts be watched, not how the watching ranks them. The present study sits in exactly that gap, asking which detection method an agency that has accepted the monitoring mandate should actually deploy, and for which anomalies. It complements our own line of hygiene work: a reproducible benchmark for hygiene anomaly detection [17] and a study of hygiene-signal augmentation for vulnerability prioritization [18], both of which treat hygiene telemetry as a first-class signal whose operational value must be measured rather than assumed.

III. Method

We model a single fleet over one snapshot and ask, for each detector and each anomaly condition, how well the detector's score ranks the anomalous hosts above the normal ones. All quantities are synthetic with documented distributions; no operational, employer, or audit telemetry is used.

A. Fleet and hygiene vector

Each seed instantiates a fleet of $N = 1{,}500$ hosts. Every host carries a 12-dimensional standardized hygiene vector $x \in \mathbb{R}^{12}$, organized as four features in each of three channels: Active Directory, endpoint, and patch. The four features within a channel are correlated, reflecting that the signals within one telemetry source move together. A normal host is drawn from a zero-mean multivariate Gaussian whose covariance Σ is block-structured: within each four-feature channel the off-diagonal correlation is $\rho = 0.70$, and across channels the features are independent. Standardizing puts every feature on a unit-variance scale so that a per-feature rule and a joint detector see the same marginal magnitudes, and any advantage a joint detector shows comes from the off-diagonal structure rather than from a scale artifact.

B. The two anomaly types

Anomalous hosts are injected at a prevalence of 8%. Each anomalous host is one of two pre-registered types.

A *single-dimension anomaly* (SDA) applies a strong shift, of mean 3.0 standard deviations, to the four features of one randomly chosen channel, leaving the other channels normal. The shifted features cross any reasonable per-feature threshold, so this anomaly is, by construction, visible to univariate inspection.

A *cross-dimension anomaly* (CDA) is a correlation-structure violation. The host's twelve features are each drawn independently from the same standard-normal marginal that normal hosts have, so every feature is individually unremarkable and crosses no per-feature threshold. What is broken is the within-channel correlation: where a normal host's four channel features move together at $\rho = 0.70$, the anomalous host's features are mutually independent. The marginal distribution is identical to normal; only the joint distribution differs. This is the canonical case of an anomaly that is invisible to per-feature inspection and visible only to a detector that models the joint structure. Formally, a normal host is drawn from $\mathcal{N}(0, \Sigma)$ with the block-correlated Σ above, while a CDA host is drawn from $\mathcal{N}(0, I)$ with the same unit marginals but no within-channel correlation, so the two differ only in their off-diagonal covariance.

Three evaluation conditions run on the same seeds and the same normal hosts: SDA-only, CDA-only, and a 50/50 mixture in which half the anomalies are SDA and half are CDA. The mixture is the realistic operational case, in which a monitor faces both obvious spikes and subtle correlation breaks at once.

C. Detectors

Every detector outputs a continuous per-host anomaly score; evaluation ranks hosts by that score, so no detector benefits from a threshold. Six detectors compete.

Rule-Max is a per-feature rule: the host's score is the maximum robust z-score across its twelve features, where the robust z uses the median and the median absolute deviation. It fires on any single feature that deviates hard, which is exactly the SDA case.

Rule-Count is a structure-blind aggregate: the score is the number of features whose robust z-score exceeds 2.5. It aggregates marginal alarms but, like Rule-Max, never inspects the joint structure, and it serves to test whether aggregation alone, short of true joint modeling, recovers the cross-channel signal.

Mahalanobis is the joint parametric detector: the score is the Mahalanobis distance of the host under a covariance $\hat{\Sigma}$ fitted to the fleet,

$$s_{\mathrm{Mah}}(x) = \sqrt{(x - \hat{\mu})^{\top} \hat{\Sigma}^{-1} (x - \hat{\mu})}, \tag{1}$$

estimated robustly [12] so that the fitted covariance is not corrupted by the anomalies it scores. The inverse covariance makes this score large for a host whose off-diagonal structure departs from normal even when its marginals do not, which is the CDA signature.

The remaining three are the standard joint detectors at library defaults: *Isolation Forest* [10] with 200 trees, scoring by the inverse of the expected isolation depth; *One-Class SVM* [9] with an RBF kernel and $\nu = 0.1$, scoring by signed distance from the fitted boundary; and *Local Outlier Factor* [8] with 20 neighbors, scoring by the local reachability-density ratio. No hyperparameter is tuned on any seed; the defaults are fixed in advance, which is the conservative choice for the joint detectors because it denies them any per-condition advantage.

D. Evaluation

For a detector that ranks the N hosts by score, average precision is the area under the precision-recall curve, equivalently the mean precision evaluated at each true-anomaly rank,

$$\mathrm{AP} = \sum_k \left(R_k - R_{k-1}\right) P_k, \tag{2}$$

where P_k and R_k are the precision and recall at the k-th threshold of the ranking. We use average precision [4] as the primary metric because the anomalies are rare and the metric is threshold-free. For each detector and condition we compute average precision on each of 25 seeds and report the mean over seeds, with comparisons decided by BCa bootstrap intervals as described next.

IV. Theoretical Analysis

The empirical ordering in Table I is not an accident of the simulation; it follows in closed form from the geometry of the cross-channel anomaly. This section derives why a correlation-violating anomaly is invisible to per-feature rules and to axis-aligned detectors yet visible to a covariance-aware one, and we match each prediction to the frozen cross-channel numbers.

We fix notation. A normal host is the 12-dimensional Gaussian $x \sim \mathcal{N}(0, \Sigma)$, where Σ is block-diagonal with three 4×4 equicorrelation blocks, unit diagonal and off-diagonal $\rho = 0.70$, and zero across channels. A cross-dimension anomaly (CDA) is $x \sim \mathcal{N}(0, I)$: the same unit marginals, but the within-channel correlation removed, so the covariance moves toward the identity. The two laws differ only in their off-diagonal covariance. The prevalence of anomalies is $\pi = 0.08$.

A. Per-feature statistics sit at the prevalence floor

Let $T(x) = g(x_j)$ be any statistic that reads a single coordinate x_j, and more generally any per-feature rule that combines the coordinates through their marginals alone, of which Rule-Max $\max_j |z_j|$ and Rule-Count $\sum_j \mathbf{1}[|z_j| > 2.5]$ are the two instances we evaluate. The j-th marginal of $\mathcal{N}(0, \Sigma)$ is $\mathcal{N}(0, \Sigma_{jj}) = \mathcal{N}(0, 1)$, and the j-th marginal of $\mathcal{N}(0, I)$ is also $\mathcal{N}(0, 1)$, because the CDA was built to preserve every marginal. Hence for every coordinate j,

$$x_j \mid \text{normal} \overset{d}{=} x_j \mid \text{CDA} \sim \mathcal{N}(0, 1), \tag{3}$$

and therefore any statistic that is a measurable function of the marginals alone has the same distribution under the two laws,

$$T(x) \mid \text{normal} \overset{d}{=} T(x) \mid \text{CDA}. \tag{4}$$

A detector whose score is distributed identically on normal and anomalous hosts cannot rank one above the other: its ranking of CDA hosts against normal hosts is exchangeable, so the expected precision at any recall equals the positive prevalence π, and the average precision equals $\pi = 0.08$ up to finite-sample fluctuation. This is the prevalence floor. The frozen cross-channel column confirms it: Rule-Max reaches 0.103 and Rule-Count 0.084, both adjacent to the 0.08 floor and statistically indistinguishable from a detector with no signal. *Match: Rule-Max* 0.103*, Rule-Count* 0.084*, prevalence floor* 0.08. The equality (4) is exact, not approximate: no per-feature rule, however it aggregates the marginals, can exceed the floor on the CDA condition, which is why Rule-Count's aggregation buys nothing over Rule-Max.

B. The squared Mahalanobis distance separates the two covariance structures

The covariance-aware detector scores a host by the squared Mahalanobis distance $D^2(x) = x^\top \Sigma^{-1} x$ in the metric induced by the inverse covariance. Its expectation under a host drawn from covariance C is

$$\mathbb{E}_{x \sim \mathcal{N}(0,C)}\left[x^\top \Sigma^{-1} x\right] = \mathrm{tr}\Big(\Sigma^{-1} C\Big), \tag{5}$$

the standard identity for the expectation of a Gaussian quadratic form. Under a normal host $C = \Sigma$, so

$$\mathbb{E}\big[D^2 \mid \text{normal}\big] = \mathrm{tr}(\Sigma^{-1}\Sigma) = \mathrm{tr}(I_{12}) = 12. \tag{6}$$

Under a CDA host $C = I$, so

$$\mathbb{E}\big[D^2 \mid \text{CDA}\big] = \mathrm{tr}(\Sigma^{-1}). \tag{7}$$

The two expectations differ exactly when $\mathrm{tr}(\Sigma^{-1}) \neq 12$, and they do precisely because the off-diagonal terms of Σ^{-1} are nonzero. For one 4×4 equicorrelation block with off-diagonal ρ, the eigenvalues of the block are $1 + 3\rho$ once and $1 - \rho$ three times, so the inverse block has diagonal entries

$$(\Sigma^{-1})_{ii} = \frac{1 + 2\rho}{(1 - \rho)(1 + 3\rho)} \tag{8}$$

and off-diagonal entries

$$(\Sigma^{-1})_{ij} = \frac{-\rho}{(1 - \rho)(1 + 3\rho)}, \qquad i \neq j \text{ in a block}, \tag{9}$$

which at $\rho = 0.70$ give diagonal 2.5806 and off-diagonal -0.7527. The block trace is $1/(1 + 3\rho) + 3/(1 - \rho) = 0.3226 + 10 = 10.3226$, and summing the three identical blocks,

$$\mathrm{tr}(\Sigma^{-1}) = 3\left(\frac{1}{1 + 3\rho} + \frac{3}{1 - \rho}\right) = 30.97. \tag{10}$$

The expected separation in score between the two structures is therefore

$$\mathbb{E}\big[D^2 \mid \text{CDA}\big] - \mathbb{E}\big[D^2 \mid \text{normal}\big] = \mathrm{tr}(\Sigma^{-1}) - 12 = 18.97 > 0, \tag{11}$$

a gap that is carried entirely by the off-diagonal entries (9): if Σ were diagonal, Σ^{-1} would be diagonal, $\mathrm{tr}(\Sigma^{-1})$ would equal 12, and the two expectations would coincide. The CDA host, drawn from the identity covariance, scores high on a metric built around the correlated Σ^{-1} because the cross terms $\sum_{i \neq j}(\Sigma^{-1})_{ij} x_i x_j$ are, in expectation, zero for a CDA host and negative for a normal host, so the anomaly is pushed up the ranking. This expected separation is exactly the power the covariance-aware detector has and the per-feature rules lack, and it is realized empirically: the Mahalanobis detector reaches 0.710 on the cross-channel condition, and the density-aware Local Outlier Factor, which scores a host by the sparsity of its joint neighborhood and is likewise disturbed when the channels decorrelate, reaches 0.638. *Match: Mahalanobis* 0.710*, Local Outlier Factor* 0.638. The same information argument can be stated through the Gaussian relative entropy between $\mathcal{N}(0, \Sigma)$ and $\mathcal{N}(0, I)$, which is strictly positive and a function of $\mathrm{tr}(\Sigma^{-1}) - \log\det\Sigma^{-1} - 12$ [19]: the two laws are distinguishable, and a covariance-aware test is the one that realizes the distinguishability.

C. An axis-aligned splitter has no access to the off-diagonal structure

The Isolation Forest isolates a host by recursive partitioning, where each split thresholds a *single* coordinate, $x_j < t$ versus $x_j \geq t$. Every region the forest can carve out is an axis-aligned box $\prod_j [a_j, b_j]$, and the probability mass a box receives depends only on the marginal cumulative distribution functions $\prod_j [F_j(b_j) - F_j(a_j)]$. By equation (3) those marginals are identical under the normal and CDA laws, so every axis-aligned box has the same mass under both, and the expected isolation depth of a CDA host equals that of a normal host. The forest therefore inherits the floor of (4): a tree of single-coordinate splits cannot represent the cross term $x_i x_j$ that distinguishes the two covariance structures, because no single split conditions on two coordinates jointly. The frozen number confirms the prediction: the Isolation Forest reaches 0.105 on the cross-channel condition, indistinguishable from the per-feature rules and from the prevalence floor. *Match: Isolation Forest* 0.105*, at the floor with Rule-Max* 0.103. The lesson is structural rather than incidental: covariance-blindness is built into any detector whose decision regions factor over the coordinates, and only a detector whose geometry couples coordinates, the covariance-aware Mahalanobis distance or the joint-density Local Outlier Factor, can see a correlation violation that leaves every marginal intact.

V. Experimental Design

The study is pre-registered. The four hypotheses, the decision thresholds, the anomaly-model constants, the detector settings, and the evaluation seed range were fixed in a dated protocol before any result on the evaluation seeds was inspected, so that no analytic choice could be steered by the outcome.

A. Hypotheses

H1 (primary, cross-channel advantage). On cross-dimension anomalies, the best joint detector exceeds the best per-feature rule on average precision by at least 0.10, with a BCa interval on the difference that excludes 0.10.

H2 (specificity). On single-dimension anomalies, the best joint detector does not beat the best per-feature rule by more than 0.05 average precision. The joint advantage is specific to cross-channel structure, not a general superiority.

H3 (mechanism). On cross-dimension anomalies, the covariance-aware Mahalanobis detector beats the structure-blind Rule-Count by a BCa-separated margin. The cross-channel gain requires joint modeling, not the aggregation of per-feature signals.

H4 (mixed realism). On the 50/50 mixture, the best joint detector beats the best rule on average precision, but by less than the cross-channel-only margin, because the single-channel cases are caught equally well by both.

TABLE I
Average precision by detector and anomaly condition (mean over 25 seeds).

Detector	Single	Cross	Mixture
Rule-Max (per-feature)	**0.836**	0.103	0.473
Rule-Count (exceedance count)	0.735	0.084	0.367
Mahalanobis (joint)	0.583	**0.710**	**0.702**
Isolation Forest	0.740	0.105	0.408
One-Class SVM	0.547	0.259	0.529
Local Outlier Factor	0.371	0.638	0.661

B. Protocol and statistics

Each hypothesis is evaluated over 25 evaluation seeds (frozen range 700 to 724). For every quantity we report the mean over seeds and a 95% BCa bootstrap interval with 10,000 resamples [5], [6]; the non-overlap of intervals, not a p-value, is the evidentiary standard, which avoids the multiple-comparison pitfalls of repeated significance tests [7]. The pre-registered failure criteria are explicit. If the best joint detector does not exceed the best rule by at least 0.05 average precision on cross-channel anomalies, H1 is declared null and the characterization is reported. If a rule baseline beats the joint detectors on cross-channel anomalies, that counter-result is reported directly. No detector hyperparameter is tuned and no anomaly-model constant is re-drawn to reach any threshold; model development used separate seeds, and the evaluation seeds were touched once, to produce the numbers below.

VI. Results

Table I reports average precision for every detector and condition, the mean over 25 seeds. All four hypotheses are supported; we take them in turn.

Joint detection wins the subtle case (H1). On cross-channel anomalies the Mahalanobis detector reaches average precision 0.710, the best of any detector, against 0.103 for Rule-Max, the best rule. The difference is 0.607, with a BCa interval of $[0.598, 0.617]$ that lies far above the pre-registered 0.10 bar, so H1 is supported decisively. The reason is geometric: the violation is a broken correlation that leaves every feature marginally normal, so the rules cannot see it and sit near the prevalence floor, while the covariance-aware metric measures exactly the off-diagonal departure (Fig. 1). The interval clears both the 0.10 pre-registered bar and the 0.05 failure threshold with wide margin.

Rules win the obvious case (H2). On single-channel anomalies Rule-Max reaches 0.836, ahead of every joint detector, and the best joint detector here is the Isolation Forest at 0.740. The joint advantage is negative: the best joint detector trails the best rule by 0.0954, with a BCa interval of $[-0.1037, -0.0847]$ that lies entirely below zero. The joint detectors do not beat the rule by more than 0.05 on this condition; the rule beats them. H2 is supported: the joint advantage is specific to cross-channel structure and does not generalize to the single-channel case that a rule already handles. A team that deployed a joint detector everywhere would be worse off on exactly the anomalies that rules catch best.

The gain requires joint modeling, not aggregation (H3). On cross-channel anomalies the structure-blind Rule-Count reaches only 0.084 against 0.710 for Mahalanobis, a difference of 0.6264 with a BCa interval of $[0.6174, 0.6362]$ that excludes zero by a wide margin. Counting per-feature exceedances, which aggregates marginal alarms without modeling their joint structure, does not recover the cross-channel signal at all; it performs no better than the per-feature maximum. The gain is attributable to joint modeling specifically, not to any aggregation of univariate signals, so H3 is supported.

The mixed case still favors joint detection, by a smaller margin (H4). On the 50/50 mixture the Mahalanobis detector reaches 0.702, the best of any detector, against 0.473 for Rule-Max, the best rule. The difference is 0.2285, with a BCa interval of $[0.2095, 0.2498]$ that excludes zero, so the joint detector wins on the realistic mixed condition. The margin is positive but smaller than the cross-channel-only margin of 0.607, exactly as H4 predicted: the single-channel half of the mixture is caught about equally well by both kinds of detector, diluting the joint advantage that the cross-channel half supplies. H4 is supported.

The choice of joint method is decisive. The cross-channel column splits the four joint detectors sharply. The covariance-aware Mahalanobis (0.710) and the density-aware Local Outlier Factor (0.638) both see the correlation violation, because both encode geometry that a broken covariance disturbs: the Mahalanobis metric directly, and the local-density score because a host whose channels are mutually inconsistent lands in a sparse region of the joint space. The One-Class SVM is intermediate (0.259). The axis-aligned Isolation Forest, by contrast, reaches only 0.105, barely above the prevalence floor and indistinguishable from the rules, because a tree of single-feature splits cannot represent a correlation that is broken while every marginal stays normal (Fig. 2). The practical consequence is sharp: the Isolation Forest is the most popular off-the-shelf anomaly detector, and a team that reached for it as the default would measure 0.105 on the very anomalies that motivate joint detection and wrongly conclude that machine learning does not help. The right joint detector for this anomaly class is covariance- or density-aware.

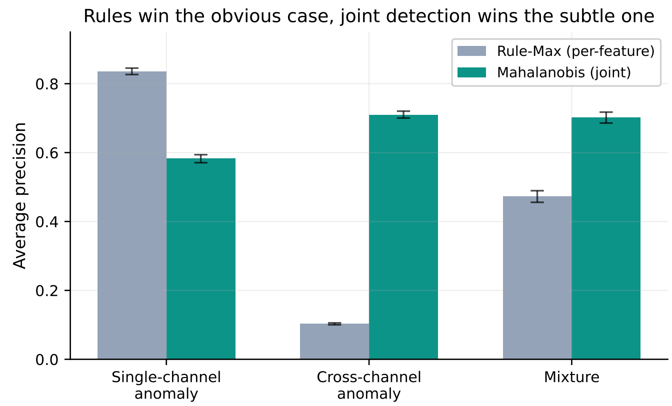

Fig. 1. Per-feature rules win on single-channel anomalies; joint detection wins on cross-channel anomalies and on the mixture.

Alt text: Grouped bar chart with average precision on the vertical axis and three anomaly conditions on the horizontal axis (single-channel, cross-channel, and mixture), each showing a Rule-Max bar beside a Mahalanobis bar with error bars. Rule-Max is taller on single-channel (about 0.84 versus 0.58), Mahalanobis is taller on cross-channel (about 0.71 versus 0.10), and Mahalanobis leads on the mixture (about 0.70 versus 0.47).

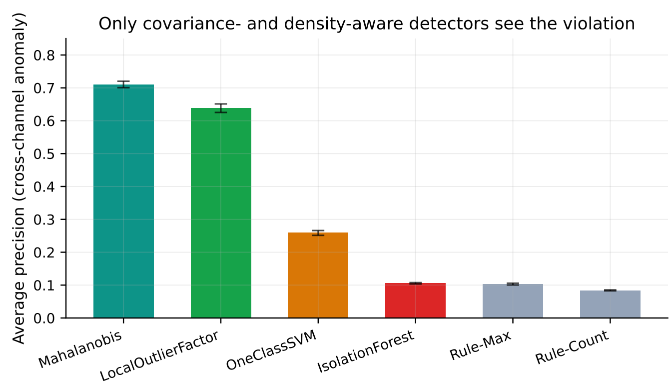

Fig. 2. On cross-channel anomalies, only covariance- and density-aware detectors see the correlation violation; the axis-aligned Isolation Forest and the per-feature rules do not.

Alt text: Bar chart of cross-channel average precision (vertical axis) for six detectors (horizontal axis) with error bars, ordered from highest to lowest: Mahalanobis (about 0.71), Local Outlier Factor (about 0.64), One-Class SVM (about 0.26), Isolation Forest (about 0.10), Rule-Max (about 0.10), and Rule-Count (about 0.08). The first two bars are far taller than the remaining four, which sit near the prevalence floor.

VII. Robustness and Sensitivity

The headline comparisons in Table I pit the single best detector against the single best rule. To check that the conclusion is robust to which detector and which condition one looks at, we report the margin of each joint detector over the *best rule* in each condition, where the best rule is the larger of Rule-Max and Rule-Count. The margin is the operationally meaningful quantity: it is what a team gains, in average precision, by adding the joint detector on top of the rule layer it already runs. Table II collects these margins, computed directly from the cell means of Table I.

The margin table sharpens three readings of the main result. First, the sign of the margin flips with the condition for the covariance- and density-aware detectors: both Mahalanobis and Local Outlier Factor trail the rule on single-channel

TABLE II
AVERAGE-PRECISION MARGIN OF EACH JOINT DETECTOR OVER THE BEST PER-FEATURE RULE, BY CONDITION. THE BEST RULE IS RULE-MAX ON EVERY CONDITION (SINGLE 0.836, CROSS 0.103, MIXTURE 0.473). POSITIVE VALUES FAVOR THE JOINT DETECTOR; NEGATIVE VALUES FAVOR THE RULE.

Joint detector	Single	Cross	Mixture
Mahalanobis	−0.253	**+0.607**	**+0.229**
Local Outlier Factor	−0.464	+0.535	+0.187
One-Class SVM	−0.289	+0.156	+0.056
Isolation Forest	−0.096	+0.002	−0.065

anomalies (margins −0.253 and −0.464) and lead it on cross-channel anomalies (margins +0.607 and +0.535), which is the conditional structure the study set out to establish, now visible across the whole detector set rather than at a single cell. Second, the Isolation Forest is the outlier among the joint detectors: its cross-channel margin is +0.002, effectively zero, and its mixture margin is negative (−0.065), so on no condition does it earn its place over the rule. An axis-aligned splitter is, by the floor argument of Section IV, the joint detector that behaves most like a rule. Third, the mixture column is positive for the three coordinate-coupling detectors and negative only for the Isolation Forest, confirming that the realistic mixed case still rewards a covariance- or density-aware second tier, at a margin between the single-channel and cross-channel extremes.

A. Adversarial reading: the correlation-only evasion

The cross-channel anomaly is not only a model of accidental misconfiguration; it is the exact shape of a deliberate evasion. Consider an adversary who has compromised a host and knows the defender monitors it with per-feature threshold rules and, possibly, an off-the-shelf axis-aligned detector. The adversary's goal is to keep the host's telemetry vector below every alarm while operating. The cheapest way to do this is to hold each feature within its normal band: keep privileged-logon counts, unsigned- binary fractions, and patch-debt figures each individually plausible, never crossing a threshold. An adversary who succeeds at this has, by construction, produced a host whose marginals are normal but whose joint structure need no longer respect the within-channel correlation that benign operation imposes, which is precisely the cross-dimension anomaly of Section IV. By equation (4) such a host is invisible to every per-feature rule: Rule-Max and Rule-Count sit at the prevalence floor (0.103 and 0.084) against this evasion because the evasion was defined to keep each marginal normal. By the box-mass argument of Section IV it is equally invisible to an axis-aligned Isolation Forest (0.105), which partitions on single coordinates and so cannot condition on the cross term the evasion disturbs. The adversary thus defeats the entire rule-based and axis-aligned monitoring stack with an evasion that costs only the discipline of staying within per-feature norms. The covariance-aware detector is the monitor that closes this gap: it scores the host by $x^\top \Sigma^{-1} x$, whose expectation (11) is raised by exactly the off-diagonal terms the evasion can no longer satisfy, and it reaches 0.710 on this anomaly class against the floor that the rules and the Isolation Forest occupy. The two-tier monitor is therefore not only an accuracy improvement on benign misconfiguration but a closure of a concrete evasion channel: an adversary who must keep every marginal normal still cannot keep the joint covariance normal, and the covariance-aware tier is the one that holds them to it. The threat we do not close is an adversary who can also forge the correlation structure itself, reproducing the within-channel covariance while hiding malicious activity; that requires controlling the joint distribution of the telemetry, a strictly harder attack than holding each marginal in band, and we note it as the residual threat the present monitor leaves open.

VIII. DISCUSSION

The question is not whether to use anomaly detection but which method to use and for which anomaly, and the results give a clean answer. The two anomaly types call for two different tools, and the mistake an operations team is most likely to make is to treat them the same.

Per-feature rules are the correct tool for single-channel anomalies. On the SDA condition Rule-Max reaches 0.836 and beats every joint detector, including the best of them by 0.0954 with an interval entirely below zero. A rule is cheap, auditable, and explainable to an authorizing official, and it loses nothing to a joint detector on the anomalies it was built for. There is no case for replacing rules on the obvious single-channel spikes; doing so adds machinery and removes interpretability for no gain.

Joint detection is decisively better only for cross-channel anomalies, the subtle compromise-like cases in which every signal is individually plausible but the host is jointly wrong. Here the best joint detector reaches 0.710 against 0.103 for the best rule, a 0.607 gap, and the structure-blind Rule-Count confirms that the gain comes from joint modeling rather than from aggregating marginal alarms: counting exceedances reaches only 0.084. This is the case that justifies the cost of a joint detector, and it is precisely the case a rule cannot reach.

Within joint detection, the standard advice to reach for an Isolation Forest is misleading for this anomaly class. Its axis-aligned random splits cannot represent a correlation violation that leaves every marginal normal, and it reaches only 0.105 on the cross-channel condition, no better than the rules it was supposed to improve on. The detectors that work are the ones whose geometry is disturbed by a broken covariance: the covariance-aware Mahalanobis (0.710) and the density-aware Local Outlier Factor (0.638). The lesson generalizes beyond this study: the right base detector is the one whose inductive bias matches the structure of the anomaly, and for a correlation violation that means a covariance- or density-aware method, not an axis-aligned one.

The deployable recommendation is a two-tier monitor. The first tier is the per-feature rules already in place, which catch the obvious single-channel spikes cheaply and remain the best tool for them. The second tier is a covariance- or density-aware joint detector that runs alongside the rules and catches the subtle cross-channel violations the rules miss. On the realistic mixture both tiers earn their place: the joint detector leads

(0.702 against 0.473), but by a smaller margin than on cross-channel anomalies alone, because the rules are still carrying the single-channel half. The two tiers are complementary, not competing, and a monitor that runs both covers both anomaly geometries at the cost of one cheap rule layer and one well-chosen joint detector.

IX. Threats to Validity

Construct validity. The cross-channel anomaly is defined by a specific synthetic mechanism, a within-channel correlation of 0.70 in normal hosts that is set to zero in anomalous ones. Real compromise telemetry may break correlation structure in other ways, by inducing spurious cross-channel correlation rather than destroying within-channel correlation, or by shifting both marginals and structure at once. The structural finding, that a correlation violation is invisible to per-feature inspection and visible to a covariance- or density-aware detector, follows from the geometry and should hold for any covariance violation, but the specific magnitudes depend on this mechanism. The single-channel shift of 3.0 standard deviations and the 8% prevalence are likewise documented priors rather than measurements.

Internal validity. The detector hyperparameters are fixed at library defaults and are not tuned on any seed (Isolation Forest 200 trees, One-Class SVM RBF with $\nu = 0.1$, Local Outlier Factor 20 neighbors). This is conservative for the joint detectors, since tuning could only raise their scores, but it means the absolute numbers for any one joint detector are untuned-default values rather than best-achievable ones. A tuned Isolation Forest might do better on the cross-channel case, though its axis-aligned construction limits how much; the comparative ordering among the joint families reflects their inductive biases, not a tuning artifact. Because the study is pre-registered and the evaluation seeds were inspected only once, the reported numbers are not the product of analytic search.

Dimensionality and generalizability. The hygiene vector has twelve dimensions in three channels of four features each. Higher-dimensional telemetry, more channels, or a different within-channel structure could change the absolute average-precision values and could affect the density-based detector in particular, since local-density estimates degrade as dimensionality grows. The fleet size of 1,500 hosts and the single-snapshot framing also abstract away temporal structure that a real monitor would exploit. We expect the qualitative finding, that joint modeling is needed for joint anomalies and that the right joint method is covariance- or density-aware, to transfer, because it rests on the geometry of the anomaly rather than on the specific constants, but absolute values should be read as illustrative of the model rather than as field estimates.

Statistical validity. Intervals are BCa bootstrap intervals over 25 seeds with 10,000 resamples [5], [6], appropriate for the small-sample, possibly skewed average-precision differences reported here. We use interval non-overlap as the evidentiary standard throughout rather than significance testing, which avoids the multiple-comparison hazards of repeated p-values across detectors and conditions [7]. Twenty-five seeds is a modest sample, but the cross-channel intervals are narrow and far from their thresholds, so the qualitative verdicts are not on the edge of significance.

X. Conclusion

Whether multivariate machine-learning anomaly detection earns its complexity over per-signal threshold rules has a conditional answer that depends on the geometry of the anomaly. On single-channel anomalies, the obvious threshold-crossing spikes, a per-feature rule wins (Rule-Max 0.836) and no joint detector improves on it, so the rule should stay. On cross-channel anomalies, the subtle compromise-like cases where every signal is individually normal but the correlation structure is broken, a joint detector is decisively better (Mahalanobis 0.710 against 0.103 for the best rule, difference 0.607), and the gain requires joint modeling rather than aggregation, since a structure-blind exceedance count reaches only 0.084. The choice of joint method is itself decisive: covariance-aware and density-aware detectors see the violation (Mahalanobis 0.710, Local Outlier Factor 0.638) while the popular axis-aligned Isolation Forest does not (0.105), so it is the wrong default for this anomaly class. On a realistic fifty-fifty mixture the joint detector still leads (0.702 against 0.473) but by a smaller margin (0.2285), because the rules still carry the single-channel half. The deployable guidance is a two-tier monitor: keep cheap per-feature rules for the obvious single-channel spikes and add a covariance- or density-aware detector for the subtle cross-channel violations. Validating the synthetic correlation model against real compromise telemetry is the natural next step toward turning these structural findings into field estimates.

Data Availability Statement

The data and code that support the findings of this study are openly available in the research-papers repository (anonymized for double-blind review; the link is provided upon acceptance). The manuscript sources, figures, analysis code, and frozen result tables are included and regenerate every reported number deterministically from the fixed seeds.

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